

Marco Cantamessa

A long voyage, from Aeneas to Elon Musk

**XXII Cycle - Alta Scuola Politecnica
Winter School on "Dynamics of Innovation"**

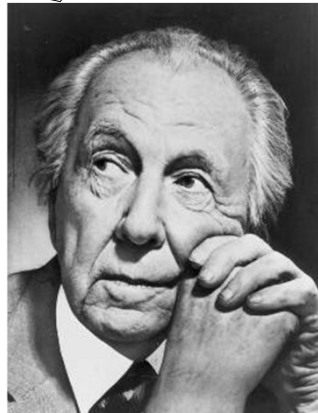
WHAT ARE WE GOING TO TALK ABOUT?

INNOVATION = "NOVUS" + "ACTION"?

IS IT ABOUT NEW IDEAS?

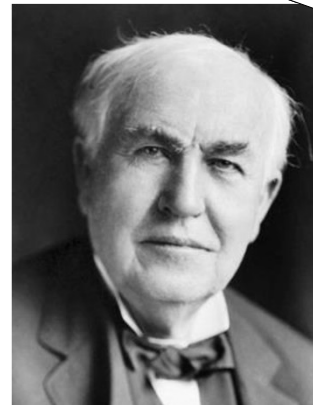
Yes, but....

**"An idea is
salvation by
imagination"**



**Frank Lloyd Wright,
1867-1959**

**"The value of an
idea lies in the
using of it"**



**Thomas Alva
Edison, 1847-1931**

INNOVATION = "NOVUS" + "ACTION"?

FOR WHOM IS IT?

"I will study hard because I want to become a great professional and be recognized for my worth"

Anonymous (and unsuccessful) ASP applicant, IX Cycle

"I will study hard because I want to make a difference in the world"

Anonymous (and successful) ASP applicant, IX Cycle

WHAT "DIFFERENCES" ARE BEING MADE IN THE WORLD, CURRENTLY?

It's not about a
specific technology or
industry

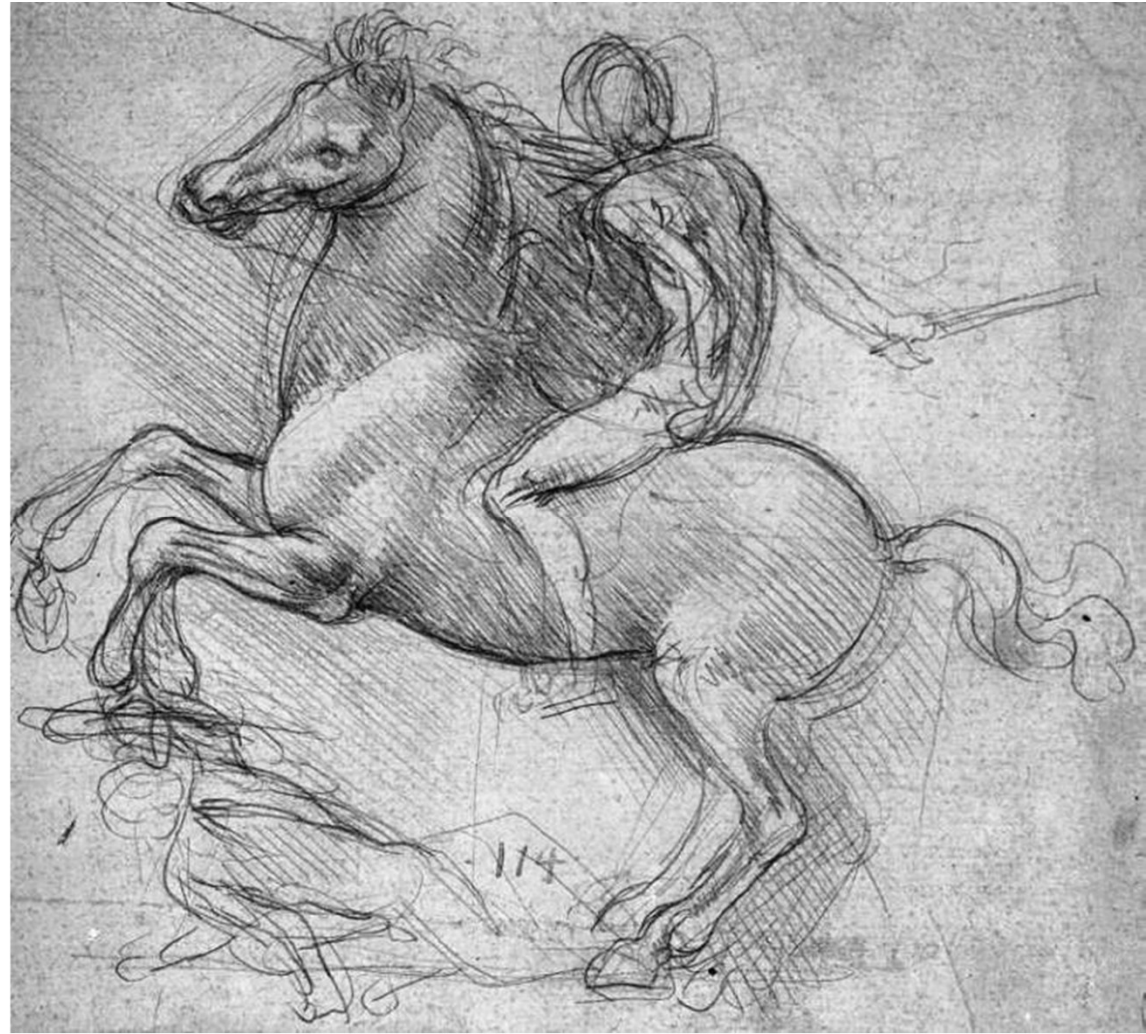
It's about the present
shaping the future

It's about "futuro" vs.
"avvenire"



Can we learn how to ride "futuro",
instead of being overrun by
"avvenire"?

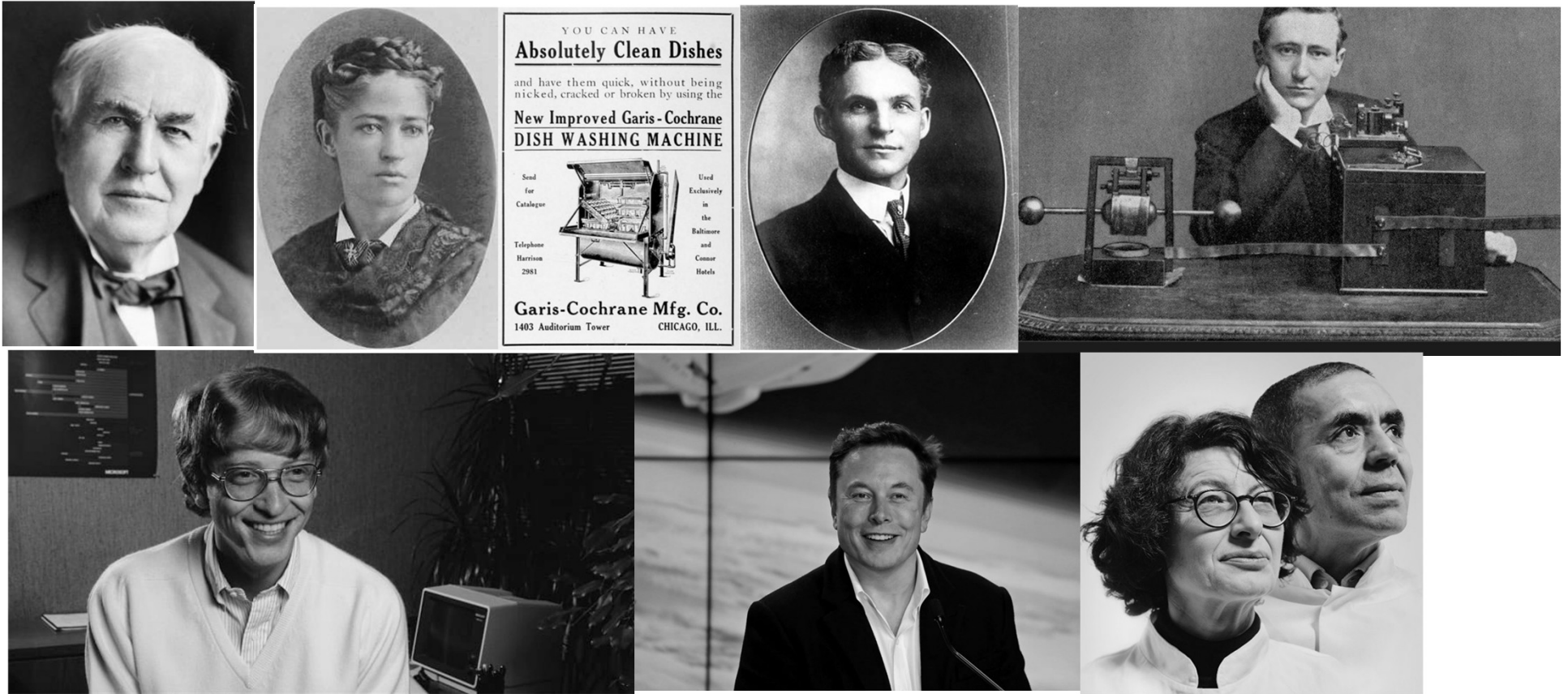
We can rationalize the past, and learn
from "rhymes of history"



Leonardo da Vinci, 1482

Studio per il monumento equestre di Francesco Sforza

(Windsor, The Royal Collection)



It's a long story of brilliant people (Schumpeter)

Thomas Edison to Josephine Cochrane, Henry Ford to Guglielmo Marconi,
Bill Gates to Elon Musk and to Ugur Sahin and Ozlem Tureci

Or is it a long story of cultures that support innovation (Mokyr)?
If so, could it be something with deeper roots?



Peter Paul Rubens, 1636
Prometheus
 (Museo del Prado, Madrid)



Unknown artist – 180-190 AD

Prometheus and Athena model the first man
 (Museo del Prado, Madrid)

What is the meaning of this?

It's about

- techne vs. physis
 - poiesis vs. sophia
- ... but there's probably more

Virgil's Aeneid, VIII book

- Aeneas has fled Troy, tries to settle in Latium and ends up in war with the locals
- Venus asks Vulcan to build new weapons for Aeneas and seduces him into building them
- Vulcan accepts and then heads for his workshop in the Aeolian islands to build invincible weapons
- Aeneas' shield bears scenes of the future of Rome, which he does not understand. But he enjoys looking at them and, unaware of the meaning, "*bears the fate of future generations on his shoulders*"... and, of course, he wins his battles

A. Van Dyck, 1632

Venus asking Vulcan for the Armour of Aeneas
(Musée du Louvre, Paris)



What is the message?

Is Virgil somehow talking about innovation, obviously ahead of time?

yes

Aeneas is engaged in

war, competition

Venus gets involved with

love, passion

Vulcan applies

technical competence

and the outcome determines

the future, short and long term

Take away any of the three, and

You can contrast the story with a similar story, on Thetis asking Hephaeustus to build Achilles' shield (Iliad, XVIII), and with a modern version by W.H. Auden... try understanding the differences!

A. Van Dyck, 1632

Venus asking Vulcan for the Armour of Aeneas
(Musée du Louvre, Paris)



**OK. Innovation is something that has to do with civilization since its very beginnings.
How does it happen?**

Fast forward to the late XIX Century

Tricky «trivial pursuit» question

- What is the profession of these people?
- Who would you want to be?
- Who were the innovators and how did they do it?
- What's the value of technology without someone managing innovation?



Who invented the light bulb?



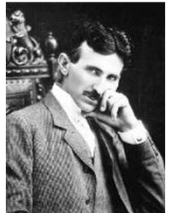
Sir Humphry Davy
(1778-1829)

Discovered the electric arc



Thomas Alva Edison
(1847-1931)

Improved and commercialized
the incandescent light bulb and
DC systems



Nikola Tesla
(1856-1943)

Invented the electric transformer,
enabling AC systems



Joseph Wilson Swan
(1828-1914)

Invented the incandescent light bulb



Alessandro Cruto
(1847-1908)

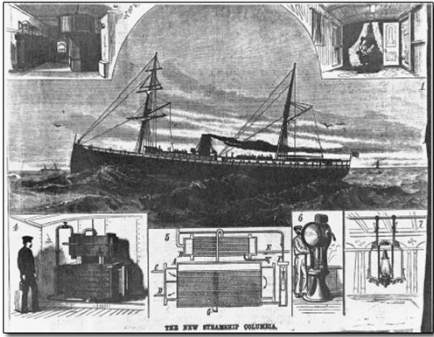
Roughly followed the same path
as Edison, but in Italy



George Westinghouse
(1846-1914)

Using inventions from Tesla and Galileo Ferraris,
developed AC systems and prevailed over Edison

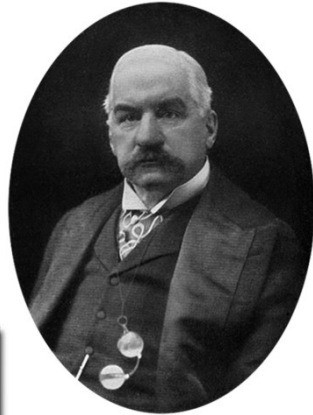
How did Edison do it?



S.S. Columbia

May 1880

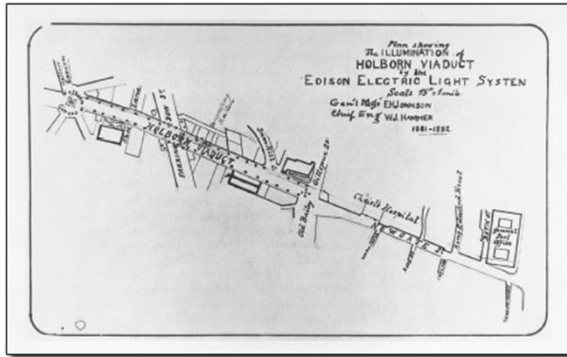
A beachhead B2B market, but he didn't stop there



J.P. Morgan's home, New York

June 1880

A niche B2C market (with some misfortunes)



Holborn Viaduct, London

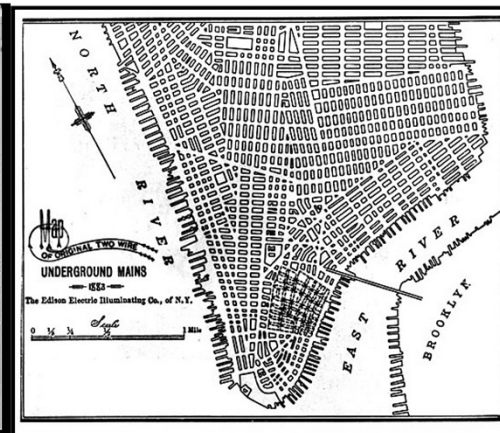
January 1882

A B2G market

Pearl Street Station, New York

September 1882

The real market, B2G+C



What can you learn from this?

Things that have to do with Edison as a person

.... technical skills, resilience, strategic vision, ability to execute

Things that have to do with business aspects

... market and stakeholder identification and development, systemic vision of technology and complementary assets, finance

A few definitions

The five Ws of (technological?) innovation	
What	• "<i>The economic exploitation of an invention</i>" (Roberts, 1998) • "<i>The act or process of introducing new ideas, devices, or methods</i>" (Merriam Webster dictionary)
Who	• Supply → People or organizations who <i>propose</i> the innovation (taking risks) • Demand → People or organizations who <i>adopt</i> the innovation (taking risks)
Why	• Craftsmanship and ingenuity making it <i>possible</i> (= technology?) • Competition making it <i>necessary</i> • A culture making change <i>agreeable</i>
When	• Cost of <i>change</i> < Cost of <i>staying still</i>
Where	• Products, processes, organizations, business models, society • Places (national/regional innovation systems, entrepreneurial ecosystems, firms) where • Resistance to change and incumbency are weaker or manageable (Juma 2016) • A 'dynamic' and 'modern' culture prevails (Phelps 2013)

Modern culture, according to Edmund Phelps (2013)

Modern economies have witnessed "Rostovian take offs" thanks to "economic dynamism", based on "flourishing" systems of "indigenous" innovation



Societies valuing 'modern' culture

- Jeffersonian individual pursuit of happiness
- Dynamism and radical change vs. tradition
- Dealing with Knightian uncertainty
- Accepting failure *and* success
- Innovation vs. trade

Economic knowledge (supply and demand side)

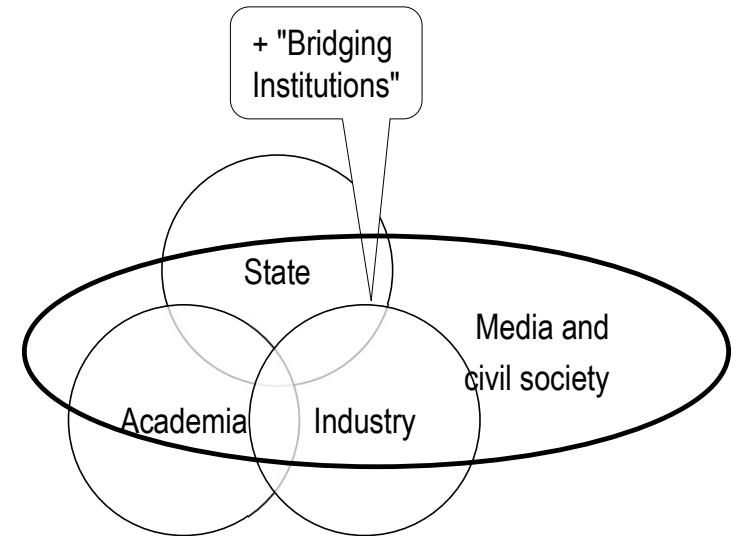
- Technical creativity
- Entrepreneurialism and strategic insight
- Technical and managerial capability to execute at scale
- Financial judgement

Institutions that

- Allow capitalism and competition (interpreted as Hayekian "discovery process") to prevail over corporativism and socialism
 - Facilitate business and do not attempt to direct or co-manage it
 - Place incentives for long-term value creation vs. short-term opportunistic behavior
 - Ensure a level playing field, support cooperation among players, and act as market players themselves
- What is the relationship between this vision and
 - traditional industrial policy (spending plans, fostering national champions)?
 - the "entrepreneurial state" model (government leading business into "grand challenges")?
 - China's current economic transformation?
 - How would you describe your own country with respect to these features of economic dynamism?
 - Any differences between regions and cities within your country?

The geography of innovation

- Local (i.e., national or regional) systems of innovation exhibit
 - A Marshallian district-like behavior, based on economies of agglomeration (specialization, resource pooling, low transaction costs, knowledge spillovers)
 - A knowledge-producing and sharing interplay between heterogeneous actors (from Etzkowitz's triple helix to Carayannis' quadruple helix)
- Recent "entrepreneurial ecosystems" are centered around (Autio et al., 2018)
 - high-growth opportunity discovery, pursuit and scaling-up
 - digital and business model innovation
 - a community of (serial) entrepreneurs and complementary actors
 - a common "economic knowledge" and culture
 - continual recycling of resources



Do these two models exhibit different features, with respect to path dependency and replicability?

So, is innovation something positive?

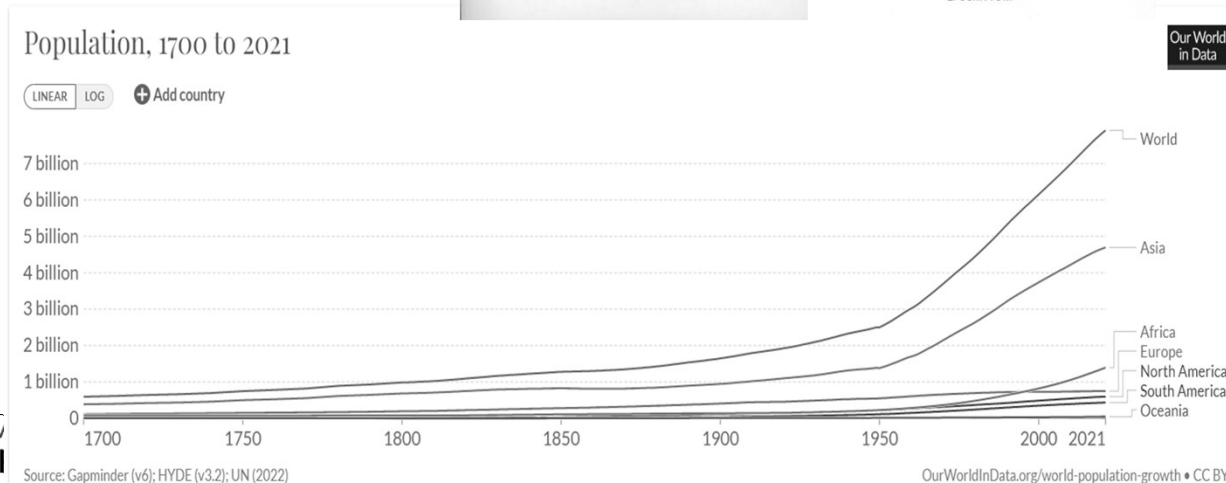
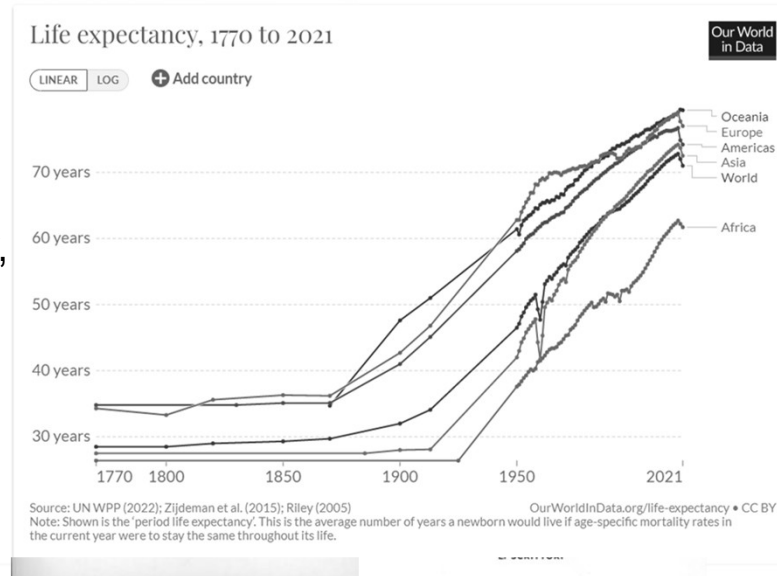
Is innovation tied to progress?

Innovation is associated to progress

How did people (even rich people!) live up to a few decades ago?

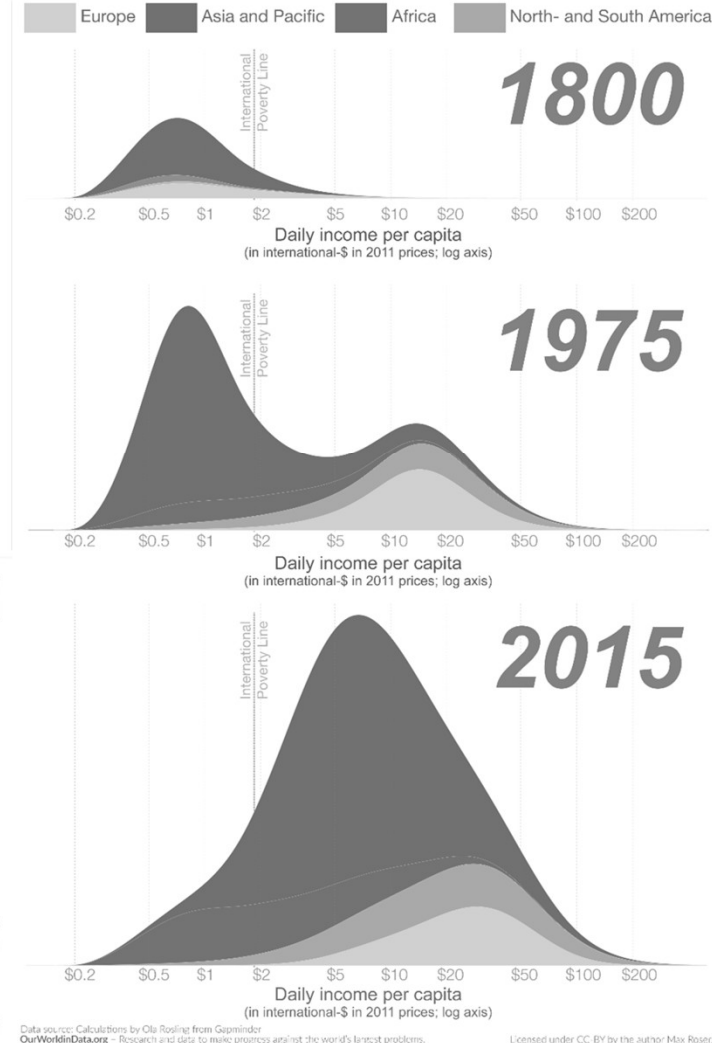
Just about any indicator of human well-being has improved substantially, globally, for a growing population

Is innovation to be credited for "pushing the boundaries" of human well-being?



Global income distribution in 1800, 1975, and 2015

Income is measured by adjusting for price changes over time (inflation) and for price differences between countries (purchasing power parity (PPP) adjustment). These estimates are based on reconstructed National Accounts and within-country inequality measures. Non-market income (e.g. through home production such as subsistence farming) is taken into account. The International Poverty Line is set by the United Nations and is the poverty line that defines extreme poverty.



So, is innovation always something cool and positive?

Not necessarily so

Innovation can have a dark side to it



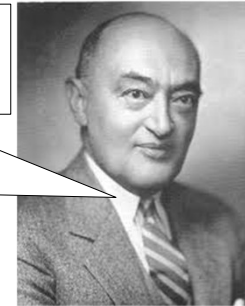
Impact of technology	Used to be	Now is becoming
Socioeconomic impact	Workers in disrupted industries losing their jobs Blue-collar work displaced by automation	White-collar work displaced by AI Gig economy (e.g., "uberized" self-employed people, engaged in perfect competition, controlled by algorithms) Decreasing labor share + wealth / income inequality (Where exactly?) What is the role of monopolies?) Deep adoption and usage divides within society
Environmental impact	Technology mitigates a hostile nature's impact on humankind	Technology can have major and irreversible impact on nature
Societal impact	Technology is shaped by society and its priorities	People are passively <i>identified</i> by technology
Ethical impact	Technology coevolves with social / cultural norms, as means to ends	Absent social norms, technology becomes the end and obeys to endogenous values (i.e., "it works")

Those who engage in innovation must care about externalities and ethical issues...

- from the "ethics of intention" to the "ethics of responsibility"
- technology determinism vs. social constructivism
- dealing with the "Collingridge dilemma" (when exactly should you regulate technology?)
- business impact, from legal liability to stakeholders' pushback

How can you study it?

Joseph Schumpeter
1883-1950



Who should study this "*perennial gale of creative destruction*"? Historians? Technologists? Economists?

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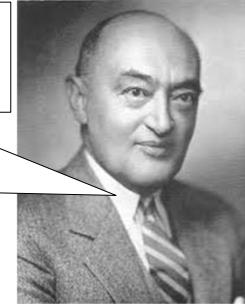
Topics	Lecturers
Innovation and economics	Marco Magnani
Innovation and management	Marco Cantamessa
Financing of innovation	Elisa Ughetto
Intellectual property rights	Giuseppe Scellato

Testimonials
Francesco Monti (Podium Technologies)
Milica Pavlovic (GSK)

You will be assigned a paper to write
In May, you will apply all this to your ASP projects!

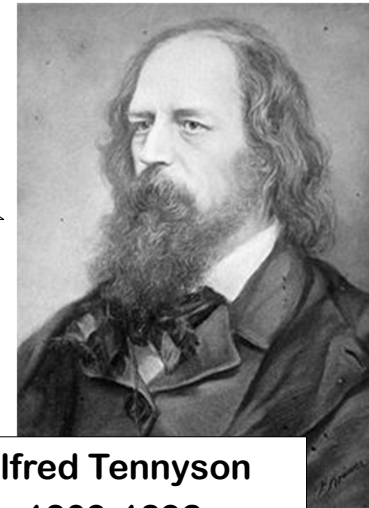
How can you study it?

Joseph Schumpeter
1883-1950



Who should study this "*perennial gale of creative destruction*"? Historians? Technologists? Economists?

Or maybe artists and poets?
Tonight, try googling Alfred Tennyson's "Ulysses"
Read it out aloud, on your own
Highlight the verses that moved you the most, and keep them in your mind throughout this school



Alfred Tennyson
1809-1892